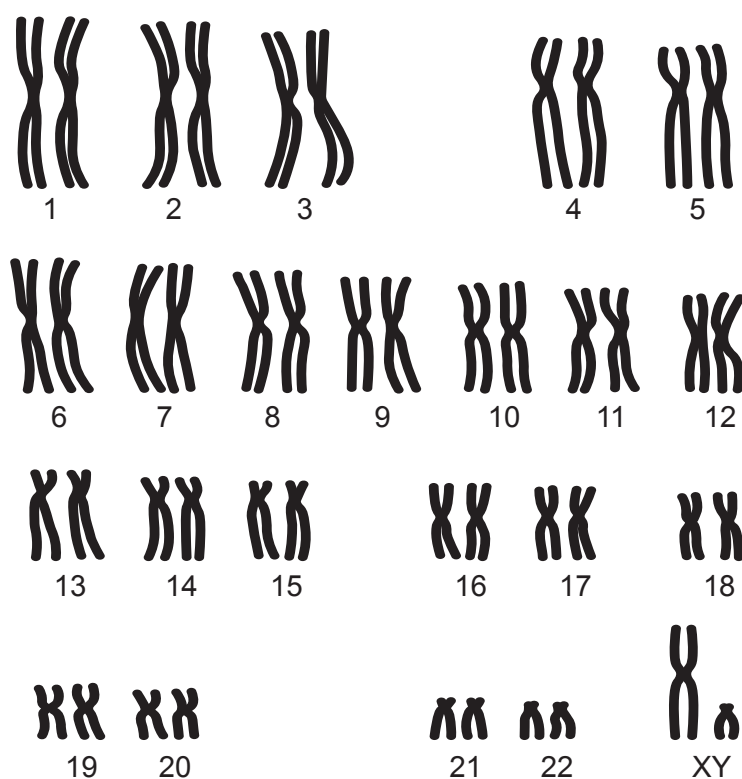


7. **Image 7.1** shows a set of chromosomes from a human male body cell.

Image 7.1



- (a) (i) State the number of chromosomes in a human body cell. [1]
-
- (ii) Draw a **circle** around the sex chromosomes in **Image 7.1**. [1]



- (b) **Images 7.2A and 7.2B** show male and female lions.

Image 7.2A



Male

Image 7.2B



Female

- (i) Complete the Punnett square below to show possible offspring when the male lion (XY) and the female lion (XX) mate. [2]

- (ii) State the expected ratio of male:female lions in the offspring. [1]

..... :

- (iii) State whether it is the male or female lion that determines the sex of the offspring. Explain your answer. [2]

.....

.....

.....

.....



(c) Lions have 38 chromosomes in their body cells.

(i) State how many chromosomes there are in a **sex cell** of a lion. [1]

.....

(ii) State the scientific name for sex cells. [1]

.....

(iii) Name the type of cell division that produces sex cells. [1]

.....

Examiner
only

10

